Math 484: Nonlinear programming

Chapter 5: Lecture 6 (Extra credit)

Posynomials

A *posynomial term* is a product $Ct_1^{\alpha_1} \cdots t_k^{\alpha_k}$, where $\alpha_1, \ldots, \alpha_k \in \mathbb{R}$ and $C > 0$. A posynomial is a sum of one or more posynomial terms.

A *constrained geometric program* is a problem of the form

$$
(GP) \quad \begin{cases} \underset{\mathbf{t} \in (0,\infty)^k}{\text{minimize}} & g(\mathbf{t}) \\ \text{subject to} & g_1(\mathbf{t}) \le 1, \\ & \ldots \\ & g_m(\mathbf{t}) \le 1, \end{cases}
$$

where $g, g_1, \ldots, g_m$ are posynomials.

An approach using KKT duality

$t_1^{-1} t_2^{-4}$

$4t_1^{-1} + 9t_2^{-1}$

$$(GP) \qquad \begin{cases} \underset{t_1, t_2 > 0}{\text{minimize}} & \dfrac{4}{t_1} + \dfrac{9}{t_2} \\ \text{subject to} & t_1 + t_2 \leq 1. \end{cases}$$

Posynomial

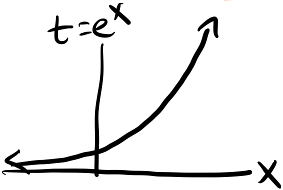
Posynomial

This problem can be solved by first arguing that $t_1 + t_2 = 1$ and then using calculus. However, this approach does not always work, and it is not always efficient. We will outline a different approach using KKT duality.

Step 1: Eliminate positivity constraints

$t_i > 0$ $t_i = e^{x_i}$, $x \in \mathbb{R}$

Replace each t_i with e^{x_i}. This is safe, because $t_i > 0$.



$$\begin{cases} \underset{t_1, t_2 > 0}{\text{minimize}} & \dfrac{4}{t_1} + \dfrac{9}{t_2} \\ \text{subject to} & t_1 + t_2 \le 1. \end{cases}$$

$$\left.\begin{cases} \underset{(x_1, x_2) \in \mathbb{R}^2}{\text{minimize}} & 4e^{-x_1} + 9e^{-x_2} \\ \text{subject to} & e^{x_1} + e^{x_2} \le 1. \end{cases}\right\} \text{These are all convex functions}$$

Now we have a convex program.

We get a convex program.

Step 2: Write each term as e^{z_j}

The next step is unintuitive. Replace each term in the previous program with e^{z_j}, using a new z_j for each term.

$$\begin{cases} \underset{(x_1,x_2)\in\mathbb{R}^2}{\text{minimize}} & 4e^{-x_1} + 9e^{-x_2} \\ \text{subject to} & e^{x_1} + e^{x_2} \le 1. \end{cases}$$

$$\begin{cases} \underset{\mathbf{z}\in\mathbb{R}^4,\mathbf{x}\in\mathbb{R}^2}{\text{minimize}} & e^{z_1} + e^{z_2} \\ \text{subject to} & e^{z_3} + e^{z_4} \le 1, \\ & e^{z_1} = 4e^{-x_1}, e^{z_2} = 9e^{-x_2} \\ & e^{z_3} = e^{x_1}, e^{z_4} = e^{x_2} \end{cases}$$

Step 3: Turn constraints into inequalities

$$\begin{cases} \underset{\mathbf{z}\in\mathbb{R}^4, \mathbf{x}\in\mathbb{R}^2}{\text{minimize}} & e^{z_1} + e^{z_2} \\ \text{subject to} & e^{z_3} + e^{z_4} \leq 1, \\ & e^{z_1} = 4e^{-x_1}, e^{z_2} = 9e^{-x_2} \\ & e^{z_3} = e^{x_1}, e^{z_4} = e^{x_2} \end{cases}$$

In minimizing the objective function, we want z_1 to be as small as possible, so z_1 will always equal any lower bound. Therefore, we can replace $e^{z_1} = 4e^{-x_1}$ with $z_1 = \log 4 - x_1$, and then with

$$z_1 \geq \log 4 - x_1.$$

Step 3: Turn constraints into inequalities

$$\begin{cases} \underset{\mathbf{z}\in\mathbb{R}^4, \mathbf{x}\in\mathbb{R}^2}{\text{minimize}} & e^{z_1} + e^{z_2} \\ \text{subject to} & e^{z_3} + e^{z_4} \le 1, \\ & z_1 \ge \log 4 - x_1,\ e^{z_2} = 9e^{-x_2} \\ & e^{z_3} = e^{x_1},\ e^{z_4} = e^{x_2} \end{cases}$$

In minimizing the objective function, we want z_2 to be as small as possible, so z_2 will always equal any lower bound. Therefore, we can replace $e^{z_2} = 9e^{-x_2}$ with $z_2 = \log 9 - x_2$, and then with

$$z_2 \ge \log 9 - x_2.$$

Step 3: Turn constraints into inequalities

$$\begin{cases} \underset{\mathbf{z}\in\mathbb{R}^4,\mathbf{x}\in\mathbb{R}^2}{\text{minimize}} & e^{z_1} + e^{z_2} \\ \text{subject to} & e^{z_3} + e^{z_4} \leq 1, \\ & z_1 \geq \log 4 - x_1, z_2 \geq \log 9 - x_2 \\ & e^{z_3} = e^{x_1}, e^{z_4} = e^{x_2} \end{cases}$$

For our constraint $e^{z_3} + e^{z_4} \leq 1$, we originally wanted $e^{x_1} + e^{x_2} \leq 1$. If $e^{z_3} \geq e^{x_1}$ and $e^{z_4} \geq e^{x_2}$ and we still have $e^{z_3} + e^{z_4} \leq 1$, then the constraint is satisfied. Therefore, we simply write

$$z_3 \geq x_1, z_4 \geq x_2.$$

Step 3: Turn constraints into inequalities

$$
\begin{cases}
\underset{\mathbf{z}\in\mathbb{R}^4,\mathbf{x}\in\mathbb{R}^2}{\text{minimize}} & e^{z_1} + e^{z_2} \\
\text{subject to} & e^{z_3} + e^{z_4} \le 1, \\
& z_1 \ge \log 4 - x_1 \\
& z_2 \ge \log 9 - x_2 \\
& z_3 \ge x_1 \\
& z_4 \ge x_2.
\end{cases}
$$

Notice that each z_i has a lower bound. This always happens when using this process.

Step 4: Put the new program in standard form

$$\begin{cases} \underset{\mathbf{z}\in\mathbb{R}^4, \mathbf{x}\in\mathbb{R}^2}{\text{minimize}} & e^{z_1} + e^{z_2} \\ \text{subject to} & e^{z_3} + e^{z_4} - 1 \leq 0, \\ & \log 4 - x_1 - z_1 \leq 0 \\ & \log 9 - x_2 - z_2 \leq 0 \\ & x_1 - z_3 \leq 0 \\ & x_2 - z_4 \leq 0 \end{cases}$$

Next, we want to solve this program using duality.

Step 5: Write the Lagrangian

$$L(\mathbf{x}, \mathbf{z}, \mu, \boldsymbol{\lambda}) = e^{z_1} + e^{z_2} + \mu(e^{z_3} + e^{z_4} - 1)$$
$$+\lambda_1(\log 4 - x_1 - z_1) + \lambda_2(\log 9 - x_2 - z_2)$$
$$+\lambda_3(x_1 - z_3) + \lambda_4(x_2 - z_4).$$

Rather than using $(\lambda_1, \ldots, \lambda_5)$, we can write $(\mu, \lambda_1, \ldots, \lambda_4)$, because we aim to get rid of μ, the coefficient of the original constraint.

Step 6: Get dual equality constraints

After some algebra,

$$L(\mathbf{x}, \mathbf{z}, \mu, \boldsymbol{\lambda}) = (e^{z_1} - \lambda_1 z_1) + (e^{z_2} - \lambda_2 z_2)$$

$$+(\mu e^{z_3} - \lambda_3 z_3) + (\mu e^{z_4} - \lambda_4 z_4)$$

$$+(-\lambda_1 + \lambda_3)x_1 + (-\lambda_2 + \lambda_4)x_2 + \lambda_1 \log 4 + \lambda_2 \log 9 - \mu.$$

Step 6: Get dual equality constraints

After some algebra,

$$L(\mathbf{x}, \mathbf{z}, \mu, \boldsymbol{\lambda}) = (e^{z_1} - \lambda_1 z_1) + (e^{z_2} - \lambda_2 z_2)$$

$$+(\mu e^{z_3} - \lambda_3 z_3) + (\mu e^{z_4} - \lambda_4 z_4)$$

$$+(-\lambda_1 + \lambda_3)x_1 + (-\lambda_2 + \lambda_4)x_2 + \lambda_1 \log 4 + \lambda_2 \log 9 - \mu.$$

Our dual objective function $h(\mu, \boldsymbol{\lambda})$ will be $-\infty$ unless

$$-\lambda_1 + \lambda_3 = 0$$

$$-\lambda_2 + \lambda_4 = 0.$$

So we fix these constraints, and we now ignore x_1, x_2.

Step 7: Compute $h(\mu, \boldsymbol{\lambda})$

$$L(\mathbf{x}, \mathbf{z}, \mu, \boldsymbol{\lambda}) = \; (e^{z_1} - \lambda_1 z_1) + (e^{z_2} - \lambda_2 z_2) + (\mu e^{z_3} - \lambda_3 z_3) \\ +(\mu e^{z_4} - \lambda_4 z_4) + \lambda_1 \log 4 + \lambda_2 \log 9 - \mu.$$

With fixed $(\mu, \boldsymbol{\lambda})$, we minimize L by letting

$$z_1 = \log \lambda_1.$$

$$z_2 = \log \lambda_2.$$

$$z_3 = \log \frac{\lambda_3}{\mu}$$

$$z_4 = \log \frac{\lambda_4}{\mu}.$$

Step 7: Compute $h(\mu, \boldsymbol{\lambda})$

$$\begin{aligned}
h(\mu, \boldsymbol{\lambda}) &= (\lambda_1 - \lambda_1 \log \lambda_1) + (\lambda_2 - \lambda_2 \log \lambda_2) \\
&+ (\lambda_3 - \lambda_3 \log \frac{\lambda_3}{\mu}) + (\lambda_4 - \lambda_4 \log \frac{\lambda_4}{\mu}) \\
&+ \lambda_1 \log 4 + \lambda_2 \log 9 - \mu \\
&= (\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4) + (\lambda_3 + \lambda_4) \log \mu - \mu \\
&+ \log \left[\left(\frac{4}{\lambda_1}\right)^{\lambda_1} \left(\frac{9}{\lambda_2}\right)^{\lambda_2} \left(\frac{1}{\lambda_3}\right)^{\lambda_3} \left(\frac{1}{\lambda_4}\right)^{\lambda_4} \right]
\end{aligned}$$

Note that the term on the last line is $\log v_0(\boldsymbol{\lambda})$, where $v_0(\boldsymbol{\lambda})$ is the GP dual objective function.

Step 8: Maximize w.r.t μ

$$
\begin{aligned}
h(\mu, \boldsymbol{\lambda}) = {} & (\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4) \\
& + (\lambda_3 + \lambda_4) \log \mu - \mu + \log v_0(\boldsymbol{\lambda})
\end{aligned}
$$

The μ terms are maximized when $\mu = \lambda_3 + \lambda_4$. Since our goal is to maximize h, we set $\mu = \lambda_3 + \lambda_4$ and forget about μ.

$$
\begin{aligned}
h(\boldsymbol{\lambda}) &= (\lambda_1 + \lambda_2) + (\lambda_3 + \lambda_4) \log(\lambda_3 + \lambda_4) + \log v_0(\boldsymbol{\lambda}) \\
&= (\lambda_1 + \lambda_2) + \log \left(v_0(\boldsymbol{\lambda})(\lambda_3 + \lambda_4)^{\lambda_3 + \lambda_4} \right).
\end{aligned}
$$

We will write $v(\boldsymbol{\lambda}) = v_0(\boldsymbol{\lambda})(\lambda_3 + \lambda_4)^{\lambda_3 + \lambda_4}$.

Step 9: Write dual to convex program

$$\begin{cases} \underset{\boldsymbol{\lambda} \geq \mathbf{0}}{\text{maximize}} & (\lambda_1 + \lambda_2) + \log v(\boldsymbol{\lambda}) \\ \text{subject to} & -\lambda_1 + \lambda_3 = 0 \\ & -\lambda_2 + \lambda_4 = 0, \end{cases}$$

where

$$v(\boldsymbol{\lambda}) = \left[\left(\frac{4}{\lambda_1}\right)^{\lambda_1} \left(\frac{9}{\lambda_2}\right)^{\lambda_2} \left(\frac{1}{\lambda_3}\right)^{\lambda_3} \left(\frac{1}{\lambda_4}\right)^{\lambda_4} (\lambda_3 + \lambda_4)^{\lambda_3 + \lambda_4} \right].$$

Step 10: Normalize $\boldsymbol{\lambda}$

Rather than consider an arbitrary $\boldsymbol{\lambda} \geq 0$, we would like to consider $\boldsymbol{\delta} \geq 0$ for which $\delta_1 + \delta_2 = 1$. We write

$$\boldsymbol{\lambda} = s\boldsymbol{\delta},$$

where $\delta_1 + \delta_2 = 1$. As

$$v(\boldsymbol{\lambda}) = \left[\left(\frac{4}{\lambda_1}\right)^{\lambda_1} \left(\frac{9}{\lambda_2}\right)^{\lambda_2} \left(\frac{1}{\lambda_3}\right)^{\lambda_3} \left(\frac{1}{\lambda_4}\right)^{\lambda_4} (\lambda_3 + \lambda_4)^{\lambda_3 + \lambda_4} \right],$$

we have

$$v(\boldsymbol{\lambda}) = v(s\,\boldsymbol{\delta}) = s^{-s} v(\boldsymbol{\delta})^s.$$

Note also $\lambda_1 + \lambda_2 = s$.

Step 10: Normalize $\boldsymbol{\lambda}$

$$\begin{cases} \underset{\boldsymbol{\delta}\geq\mathbf{0}, s\geq 0}{\text{maximize}} & s - s\log s + s\log v(\boldsymbol{\delta}) \\ \text{subject to} & -\delta_1 + \delta_3 = 0 \\ & -\delta_2 + \delta_4 = 0 \\ & \delta_1 + \delta_2 = 1 \end{cases}$$

The obj. func. is maximized when $s = v(\boldsymbol{\delta})$, giving

$$\begin{cases} \underset{\boldsymbol{\delta}\geq\mathbf{0}}{\text{maximize}} & v(\boldsymbol{\delta}) \\ \text{subject to} & -\delta_1 + \delta_3 = 0 \\ & -\delta_2 + \delta_4 = 0 \\ & \delta_1 + \delta_2 = 1 \end{cases}$$

The dual program

The final program DGP is the dual to the primal program GP.

$$(GP) \qquad \begin{cases} \underset{t_1, t_2 > 0}{\text{minimize}} & \dfrac{4}{t_1} + \dfrac{9}{t_2} \\ \text{subject to} & t_1 + t_2 \leq 1. \end{cases}$$

$$(DGP) \qquad \begin{cases} \underset{\boldsymbol{\delta} \geq \mathbf{0}}{\text{maximize}} & v(\boldsymbol{\delta}) \\ \text{subject to} & -\delta_1 + \delta_3 = 0 \\ & -\delta_2 + \delta_4 = 0 \\ & \delta_1 + \delta_2 = 1 \end{cases}$$

What do we notice?

- The primal has four terms, and the dual has four variables.
- The primal has a constraint involving terms 3 and 4, and $v(\boldsymbol{\delta})$ has an extra factor with δ_3 and δ_4.
- The exponents of t_1, in order, are $-1, 0, 1, 0$. Also, $-\delta_1 + 0\delta_2 + \delta_3 + 0\delta_4 = 0$.
- The exponents of t_2, in order, are $0, -1, 0, 1$. Also, $0\delta_1 - \delta_2 + 0\delta_3 + \delta_4 = 0$.

This is the general pattern of the dual.

Solving the dual, then the primal

The hope with the dual is always that it is computationally simpler than the primal.

With computational methods, we find that the dual has a solution $\boldsymbol{\delta}^* = (\frac{2}{5}, \frac{3}{5}, \frac{2}{5}, \frac{3}{5})$, and $v(\boldsymbol{\delta}^*) = 25$.

We will:

- Compute the sensitivity vector $(\mu^*, \boldsymbol{\lambda}^*)$ of the convex program (from Step 4) that we used to get the dual.
- Find $\mathbf{x}, \mathbf{z}$ to minimize $L(\cdot, \cdot, \mu^* \boldsymbol{\lambda}^*)$ and use this to solve the primal GP.

Getting the sensitivity vector

- We set $\boldsymbol{\lambda} = s\,\boldsymbol{\delta}$, where $s = v(\boldsymbol{\delta})$. This gives us $\boldsymbol{\lambda}^* = v(\boldsymbol{\delta}^*)\,\boldsymbol{\delta}^* = (10, 15, 10, 15)$.
- We set $\mu = \lambda_3 + \lambda_4$, meaning $\mu^* = \lambda_3^* + \lambda_4^* = 25$.

This gives us a sensitivity vector

$$(\mu^*, \boldsymbol{\lambda}^*) = (25, 10, 15, 10, 15).$$

We learn that the program

$$
\begin{cases}
\underset{\mathbf{z}\in\mathbb{R}^4, \mathbf{x}\in\mathbb{R}^2}{\text{minimize}} & e^{z_1} + e^{z_2} \\
\text{subject to} & e^{z_3} + e^{z_4} \leq 1, \\
& \log 4 - x_1 - z_1 \leq 0 \\
& \log 9 - x_2 - z_2 \leq 0 \\
& x_1 - z_3 \leq 0 \\
& x_2 - z_4 \leq 0
\end{cases}
$$

has a sensitivity vector $(\mu^*, \boldsymbol{\lambda}^*) = (25, 10, 15, 10, 15)$.

In order to solve the primal GP, we find $(\mathbf{z}^*, \mathbf{x}^*)$ to solve this convex program by minimizing $L(\cdot, \mu^*, \boldsymbol{\lambda}^*)$.

Minimizing the Lagrangian

To minimize the Lagrangian with fixed $(\mu, \boldsymbol{\lambda})$, we set

$$z_1 = \log \lambda_1.$$

$$z_2 = \log \lambda_2.$$

Using $(\mu^*, \boldsymbol{\lambda}^*) = (25, 10, 15, 10, 15)$, we get

$$(z_1^*, z_2^*) = (\log 10, \log 15).$$

(We do not care about other z_i^* or x_i^*, because we only need z_1^* and z_2^* to find t_1^*, t_2^*.)

Obtaining t_1^* and t_2^*

Finally, recall that we set $\frac{4}{t_1} = 4e^{-x_1} = e^{z_1}$. This means that

$$t_1^* = 4e^{-z_1^*} = \frac{2}{5}.$$

Similarly, we set $\frac{9}{t_2} = 9e^{-x_2} = e^{z_2}$. This means that

$$t_2^* = 9e^{-z_2^*} = \frac{3}{5}.$$

Therefore,

$$(t_1^*, t_2^*) = \left(\frac{2}{5}, \frac{3}{5}\right).$$

Other observations

We have

$$z_3^* = \log \frac{\lambda_3^*}{\mu^*} = \log \frac{\lambda_3^*}{\lambda_3^* + \lambda_4^*} = \log \frac{\delta_3^*}{\delta_3^* + \delta_4^*}.$$

This also tells us that

$$t_1^* = e^{z_3^*} = \frac{\delta_3^*}{\delta_3^* + \delta_4^*} = \frac{2}{5}.$$

Other observations

We have

$$z_4^* = \log \frac{\lambda_4^*}{\mu^*} = \log \frac{\lambda_4^*}{\lambda_3^* + \lambda_4^*} = \log \frac{\delta_4^*}{\delta_3^* + \delta_4^*}.$$

This also tells us that

$$t_2^* = e^{z_4^*} = \frac{\delta_4^*}{\delta_3^* + \delta_4^*} = \frac{3}{5}.$$

These observations indicate a more general pattern which we will cover next.